RESEARCH PAPER OVERVIEW

# Text Classification in the LLM Era: Where Do We Stand?

Do we still need to train classifiers, or can we just prompt a big model?

Vajjala & Shimangaud · National Research Council Canada / McGill · arXiv 2502.11830 (Feb 2025)

# Why this paper: gaps and questions

## Research gaps

- Most LLM classification studies use English only
- Most use proprietary models (GPT-3.5/4), few open LLMs
- Zero-shot, few-shot, synthetic data and full training are rarely compared together
- Few tests with many labels (50+ classes)

## Four research questions

1. Zero-shot LLM vs a classifier trained on all data?
2. Does few-shot fine-tuning beat zero-shot?
3. Does a synthetic-data classifier beat zero-shot?
4. Is fine-tuning an LLM the way to go?

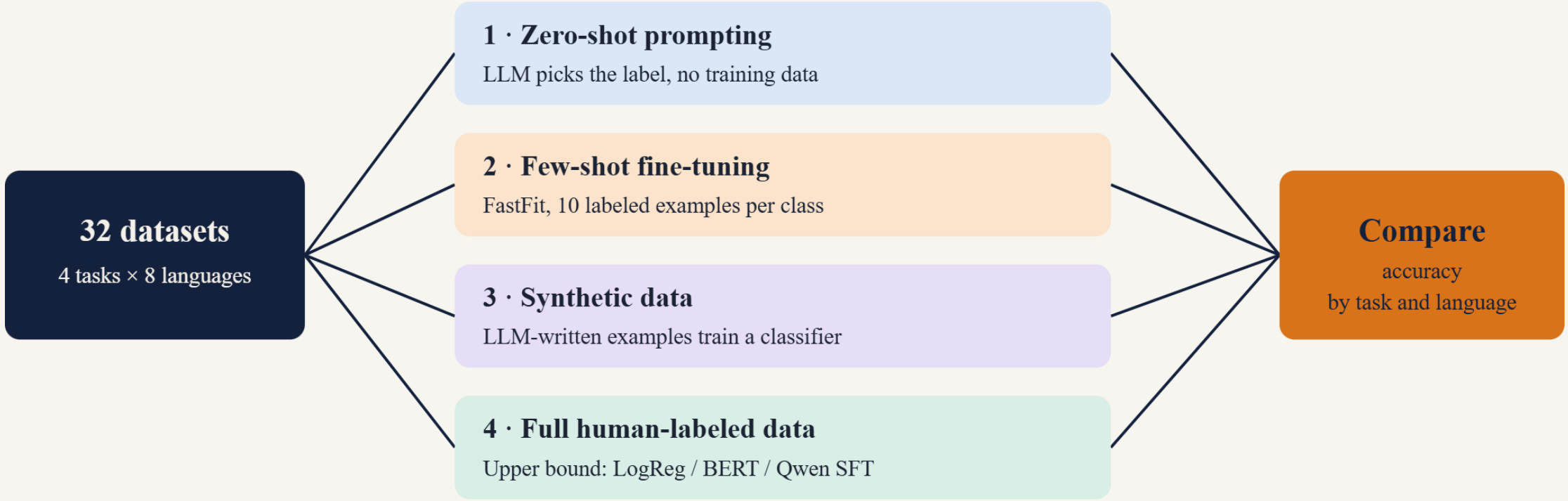
# Setup: 4 datasets × 8 languages = 32 datasets

| Dataset | Task | Classes | Text type |
|---|---|---|---|
| SENTIMENT | Positive / negative / neutral | 3 | Tweets |
| TAXI1500 | Topic of a sentence (faith, sin, grace…) | 6 | Bible sentences |
| SCENARIO | Domain of a request (music, alarm…) | 18 | Voice commands |
| INTENT | Exact user intent | 60 | Voice commands |

**Languages** Arabic · English · French · German · Hindi · Italian · Portuguese · Spanish

Metric: accuracy. Models are trained per language, on top of multilingual pre-trained models.

# Architecture: the overall study design



# How each approach works

## 1 · Zero-shot prompting

Ask an LLM to name the label. Models: GPT-4, Qwen2.5-7B, Aya-23-8B, Aya-Expanse-8B. The Instructor library forces a valid label; extra explanations count as errors.

## 2 · Few-shot (FastFit)

Only 10 labeled texts per class. Contrastive learning plus token similarity teaches the model to match texts to labels. Trains in 5–15 minutes on a laptop.

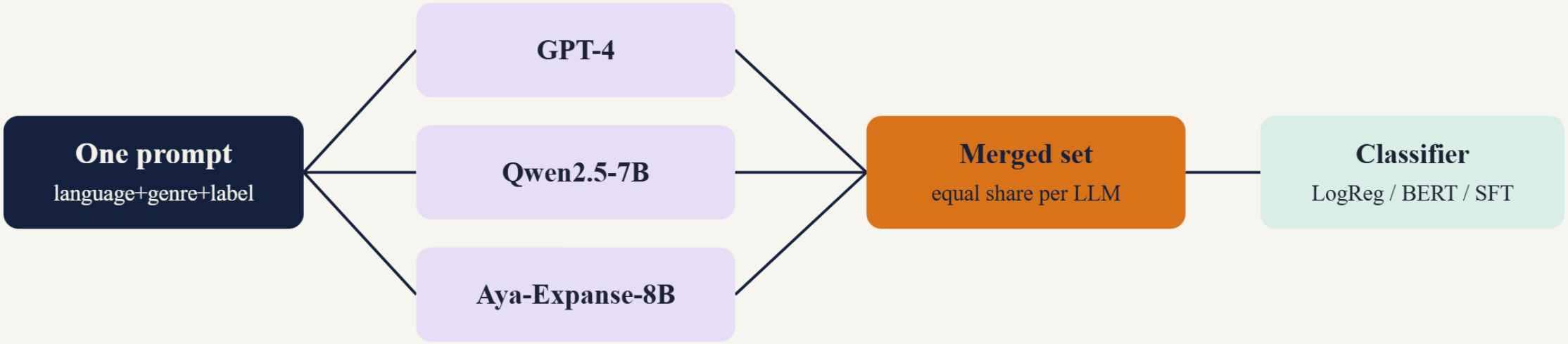
## 3 · Synthetic data

GPT-4, Qwen and Aya-Expanse each write labeled examples from one prompt. A normal classifier is then trained on that generated data only.

## 4 · Full-data classifiers

Logistic regression on gte-multilingual embeddings, fine-tuned multilingual BERT, and instruction-tuned Qwen2.5-7B (SFT). Trained on real or synthetic data.

# Architecture: synthetic data from several LLMs

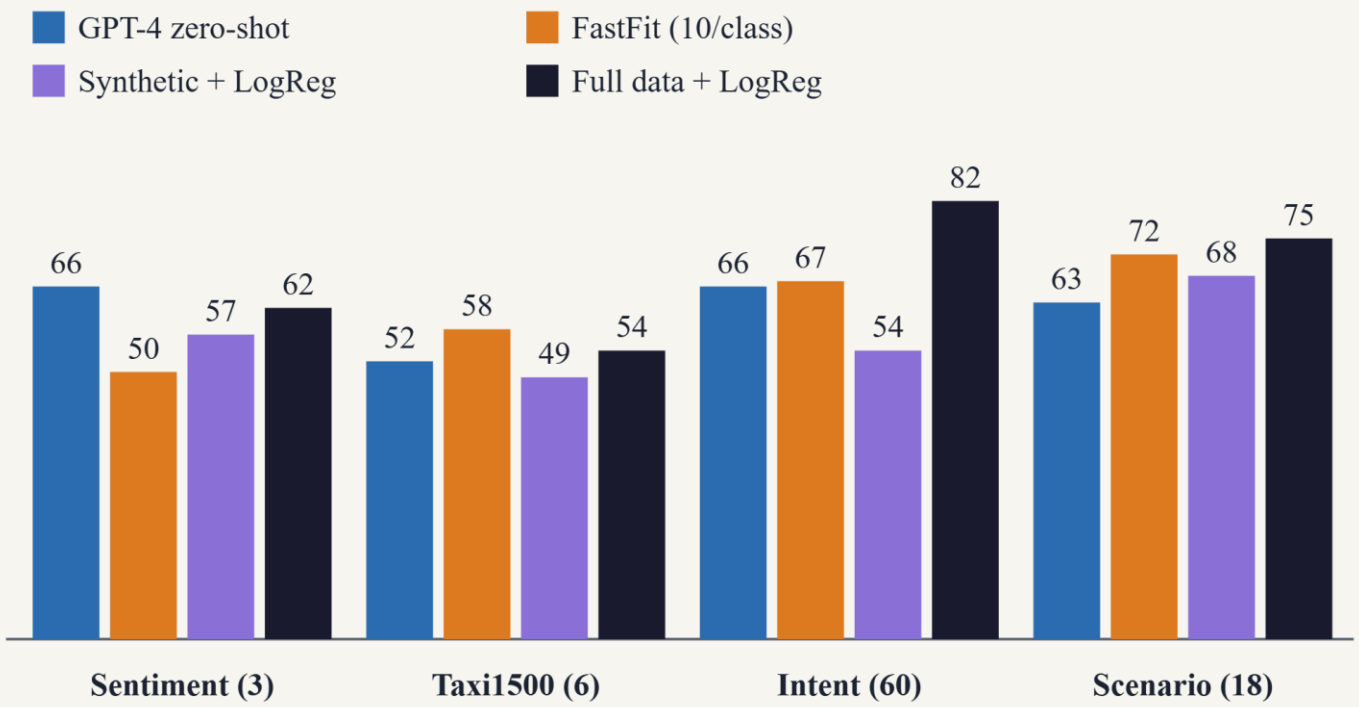



**Qwen** was the best single source with few labels (Twitter 0.51, Taxi 0.43)

**GPT-4** was steady everywhere and best with many labels (Intent 0.53)

**Mixing all three** beat any single source, and two are small local models

# Results: accuracy by approach (avg. of 8 languages)

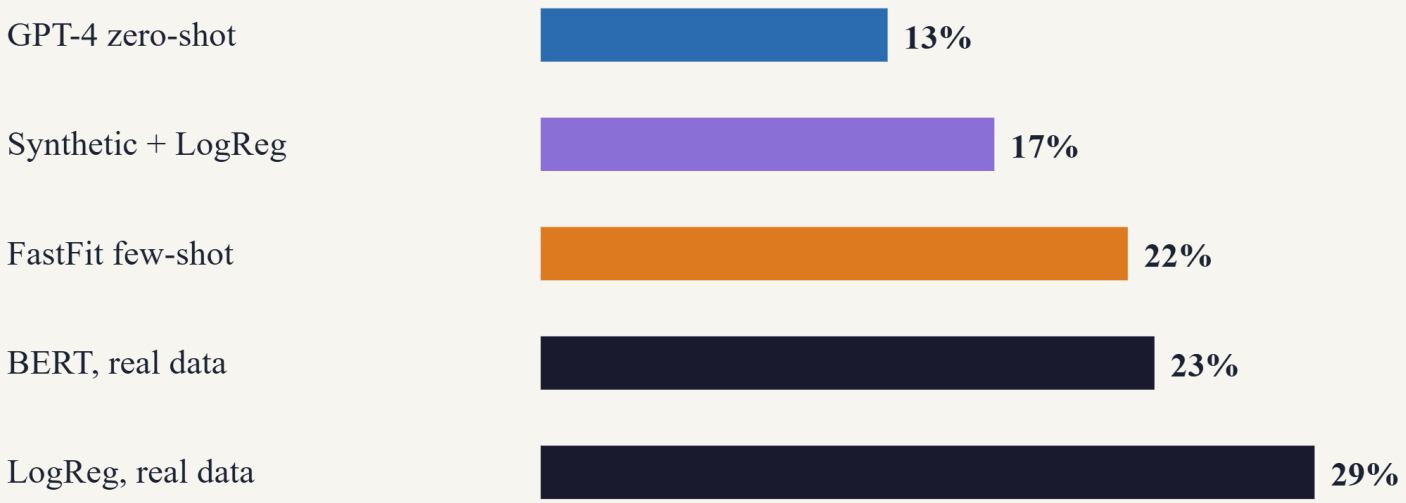



**Few classes:** zero-shot GPT-4 beats full-data training on sentiment (66 vs 62)

**Many classes:** full data wins big (Intent 82 vs 66)

**10 examples per class** match or beat GPT-4 on 3 of 4 tasks

# Language gaps: best vs worst language




Same model, big swings: BERT on Taxi1500 scores 77% in English but 35% in Italian

Arabic and Hindi are often weakest; they also cost up to 3× more tokens to generate

# Guidelines for practitioners

### 1 · Sentiment

Just prompt an LLM; it beats task-specific training.

### 2 · Little labeled data

Label about 10 per class and run FastFit; it beats zero-shot elsewhere.

### 3 · No labeled data

Generate synthetic data from several LLMs; it beats open-LLM zero-shot.

### 4 · Plenty of labeled data

Train on it. Fine-tuned LLMs win with many classes; logistic regression is a cheap strong baseline.

# Limitations and future work

## Limitations

- One prompt, short texts only
- 7 of 8 languages are Indo-European
- Open LLMs are small (7–8B) next to GPT-4
- Single runs, no quality review of synthetic text

## Future work

- More languages and language families
- Multi-label datasets
- Longer documents
- Few-shot prompting and larger open LLMs

**Bottom line: labeled data still matters; LLMs are a strong shortcut when labels are scarce.**